

Geometry Model

Version 1.0

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University of Michigan – Summer 2026

‘The Living Doc’

Big Question:

- Why does the *Xenopus* cell-free extract sometimes form compartments, and why does it sometimes not?

Hypothesis:

- **Tubulin compartment formation reorganizes the cytoplasm into a mechanically ordered state that alters intracellular transport properties, leading to faster and/or more synchronized CDK1 signaling. The synchronization is then dependent on Tubulin compartment Geometry.**

Experimental Observations:

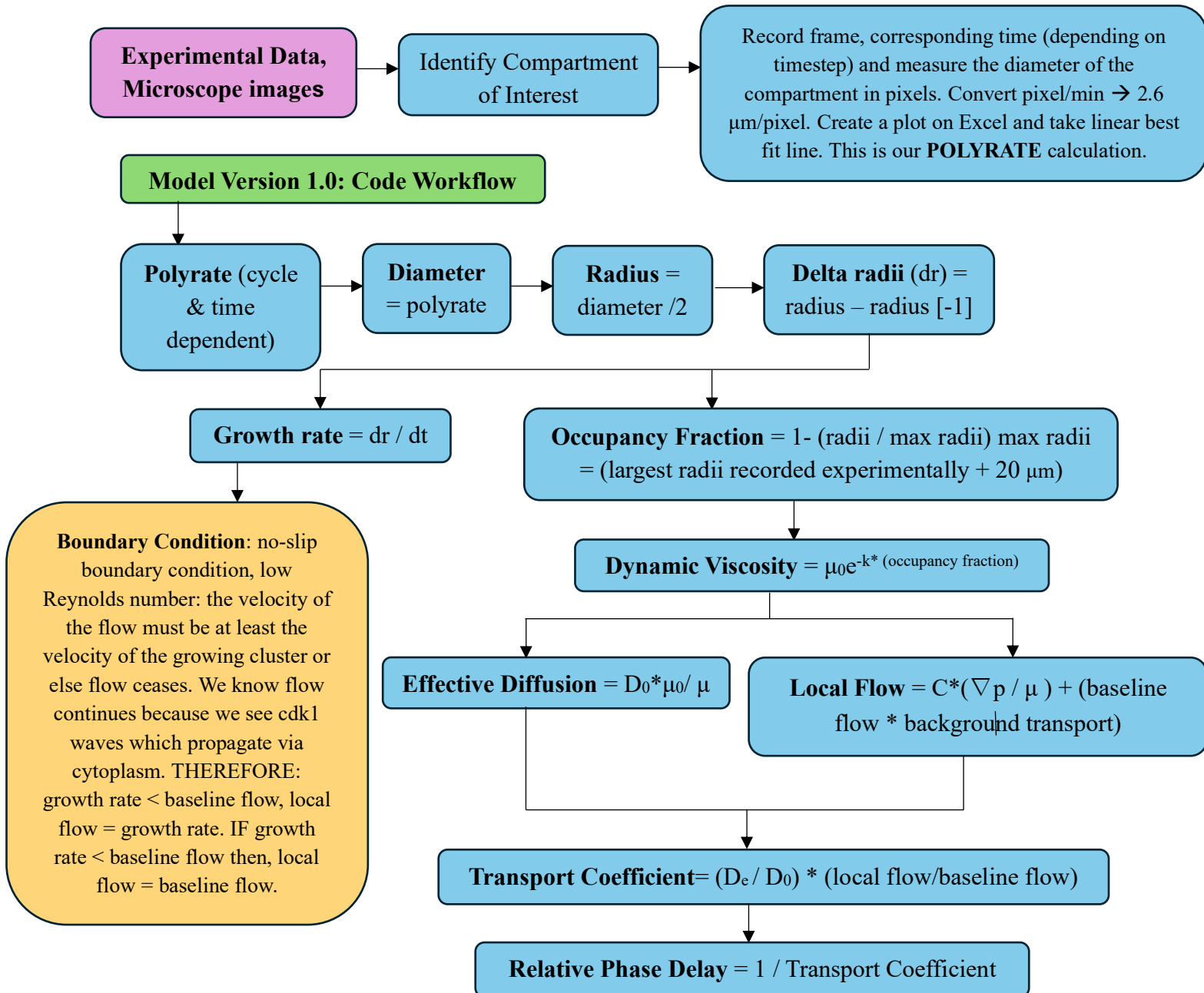
- Sometimes compartments form, sometimes they don't.
- Compartment shape changes throughout the sample's lifetime.
- Sometimes the compartments divide, sometimes they combine.
- The size of the compartment varies over time.
- If left long enough, the compartments will start to crosslink with each other across the regions of no tubulin.
- Wave speed appears to be relatively consistent but then has a very steep drop off prior to arrest.
- Periodicity does the same.
- Wave data (non-compartment forming samples) last longer than compartment forming samples. More cdk1 waves.
- Parallel vector behavior is the most common across FRET /CFP and RFP channels.
- Sometimes, mechanical activity appears to spike prior to mitotic activity.
- Kymographs show parallel wave-speeds (nearly instantaneous) in compartment forming samples until arrest. This is also seen in the wave data (non-compartment forming samples) early on, but the wave speed does slow over time. For compartments, it seems to be a bit more steep of a drop off.
- 'Highway' paths of cytoplasm are formed in compartment samples where no tubulin appears to live.
- There is a lot of flow issues in compartment data (in terms of quantifying on ImageJ).

Variables:

- Diameter $\rightarrow D(t)$, from ImageJ data $\rightarrow$ **Experimental**
- Radius $\rightarrow r$, calculated from Diameter $\rightarrow$ **Calculated**
- Growth rate $\rightarrow G_r(\Delta r)$, calculated from change in radii $\rightarrow$ **Calculated**
- Occupancy Fraction $\rightarrow O_f(r, \text{max_radii})$, calculated from current radii and max radii (assumed from ImageJ data) $\rightarrow$ **Calculated**
- Dynamic Viscosity $\rightarrow \mu(H_f)$, calculated from baseline viscosity, crowding constant and Occupancy fraction $\rightarrow$ **Calculated**

- Effective Diffusion $\rightarrow D_{\text{eff}}(\mu)$, calculated from baseline diffusion and dynamic viscosity $\rightarrow$ **Calculated**
- Local Flow $\rightarrow U_{\text{local}}(\mu)$, calculated from the pressure gradient and dynamic viscosity, inclusion of boundary condition given by growth rate $\rightarrow$ **Calculated** and **Boundary** and **Modeled**
- Transport coefficient $\rightarrow T_{\text{coe}}(U_{\text{local}}, D_{\text{eff}})$, calculated from local flow, baseline flow, baseline diffusion and effective diffusion $\rightarrow$ **Calculated** and **Modeled**
- Delta Relative Phase Delay $\rightarrow \Delta \text{Rel}_{\text{phaselag}}(T_{\text{coe}})$, calculated from the transport coefficient $\rightarrow$ **Calculated** and **Modeled** and **Comparable**

Workflow, from experimental data to validation:



Equations & Justifications:

- **Polyrate:** This is derived by recording the frame and corresponding time (time step dependent) along with the diameter change in pixel and creating a linear line of best fit. In later models this can be modified to a polynomial line of best fit for more accurate growth tracking. The polyrate is then multiplied by the pixel per micron scaling number of 2.6.
Equation: $\text{POLYRATE} = (mx + b) * 2.6$
Units: μm
- **Diameter:** Since the polyrate is essentially a linear equation for diameter given some time, Diameter can be set equal to polyrate.
Equation: $\text{DIAMETER} = \text{POLYRATE}$
Units: μm
- **Radius:** this equation is self-explanatory.
Equation: $\text{RADIUS} = \text{DIAMETER} / 2$
Units: μm
- **Δ Radii:** This is self-explanatory .
Equation: $\Delta\text{radii} = \text{Radii}[i] - \text{Radii}[i-1]$
Units: μm
- **Growth Rate:** This is the rate of the tubulin clusters growing. This value is also the velocity on the cluster at a given time.
Equation: $G_{\text{rate}} = \Delta r / \Delta t$
Units: $\mu\text{m} / \text{min}$
- **Compartment Occupancy Fraction:** This equation is assumed in Version 1 of this model. It's purpose is to calculate how large the compartment is relative to its maximum. This fraction may not exceed 1 in the version 1(though later, if a model would like to introduce a crosslinking factor amongst the clusters this will be modified). This is measured by taking the current radii of the cluster and dividing it by its maximum. Later we can use a volume calculation to align more with what other models do. This calculation will have downstream effects on viscosity. Theoretically, the larger the compartment, the less free flowing tubulin will sit in these highway regions effecting viscosity. It might also be interesting to look at the stress and pressure changes in latter models of the compartment formation.
Equation: $O_{\text{fraction}} = (\text{radii} / \text{max radii})$
Units: unitless
- **Dynamic Viscosity:** This equation is also currently assumed in Version 1 of this model. The equation is taken from basic fluid mechanics where the baseline viscosity is multiplied and exponential value of the occupancy fraction. The reasoning here is that the viscosity will change along the highway as more tubulin polymerize into the clusters. The baseline viscosity and the crowding constant, k, put as 1 to minimize noise. The

exponential is taken as – because as the occupancy fraction increases, viscosity along said highway should decrease.

Equation: $\mu = \mu_0 e^{-k\phi}$

Units: Pa * min

- **Effective Diffusion:** The effective diffusion parameter should be based on the current diffusion and the viscosity. This equation was derived from basic fluid mechanics and the relation between diffusion and viscosity (inverse). The dynamic viscosity is normalized to it's baseline.

Equation: $D_{\text{eff}} = (\mu_0 / \mu) * D_0$

Units: $\mu\text{m}^2 / \text{min}$

- **Local Flow:** Local flow was derived using a constitutive relationship motivated by low-Reynolds-number transport. Lower local viscosity produces higher local flow. A lower boundary is imposed by the compartment growth velocity. C is a proportionality constant to ensure that the parameter is in the correct units.

Equation: $U_{\text{local}} = C(\nabla p / \mu) + (U_{\text{baseline}} * \text{Transport}_{\text{background}})$

Units: $\mu\text{m} / \text{min}$

- **Transport Coefficient:** This value is based on the relationship between diffusion and flow to determine transport. This equation is also a simplification unique to Version 1 of the model. The diffusion and flow are normalized to the baselines.

Equation: $T_{\text{coe}} = (D_e / D_0) * (U_{\text{local}} / U_{\text{baseline}})$

Units: unitless

- **Relative Phase Delay:** This equation was derived based on the inverse relationship between transport and how much the cdk1 activity could lag relative to the mechanical. Since if the transport increases, the lag between the large mechanical activity spike and the mitotic activity spike would decrease since information is being passed 'quicker'.

Equation: $\Delta P_{\text{lag}} = 1 / T_{\text{coe}}$

Units: unitless

Assumptions:

- Baseline flow = 1.0
- A background transport variant that accounts for ATP consumption, microtubule density, cytoplasmic reorganization, ect.

$$\text{background_transport} = 1.0 + 0.15 * \text{np.sin}(2 * \text{np.pi} * i / (6 * T))$$

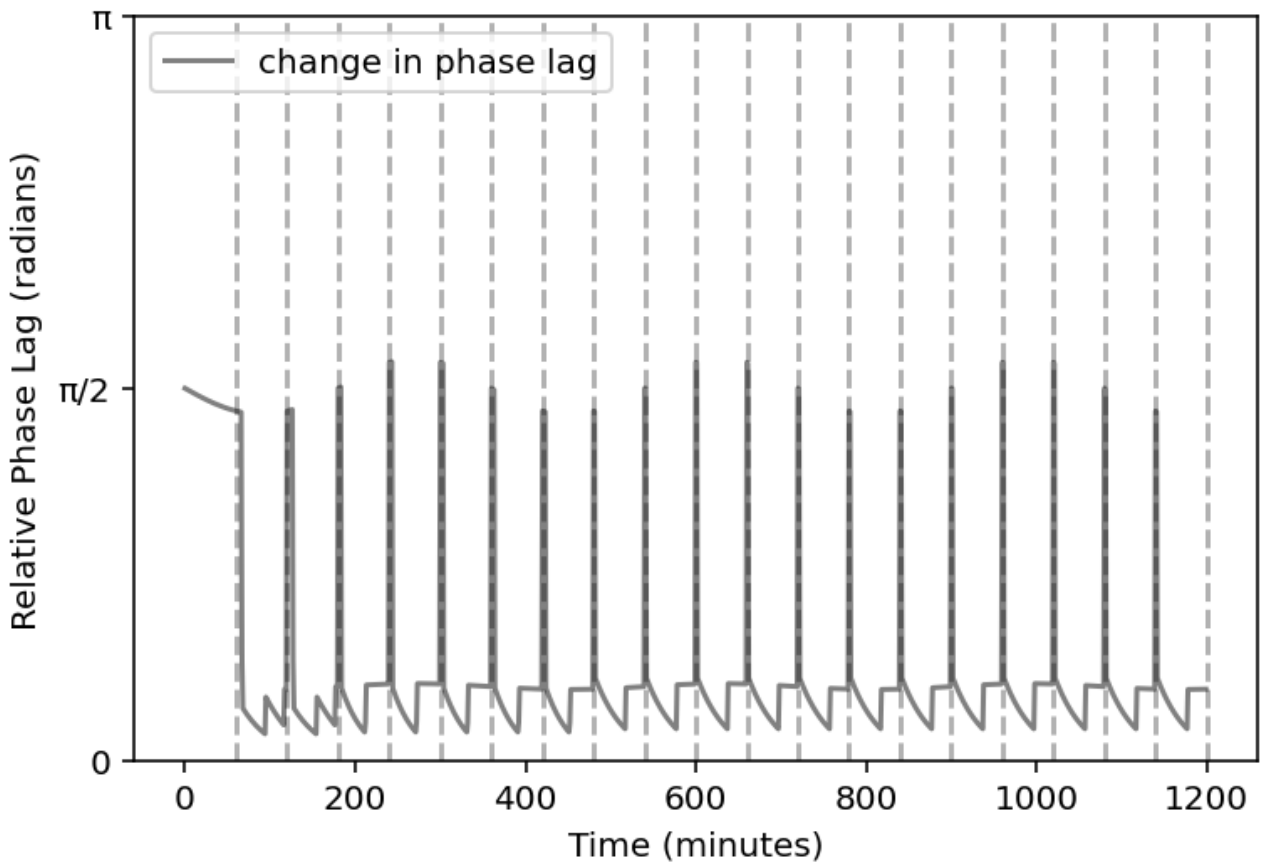
- Crowding constant, $k = 1.0$
- Baseline Diffusion = 1.0
- Pressure Gradient = 1.0
- Treating the cytoplasm and a **Low Reynolds Number**, this implies that the fluid is dominated by viscosity and that there is little to no inertia → no momentum. The justification for this is that the flow appears to be laminar after the initial compartment formation. This can be validated later when we get viscosity, density and linear size data.

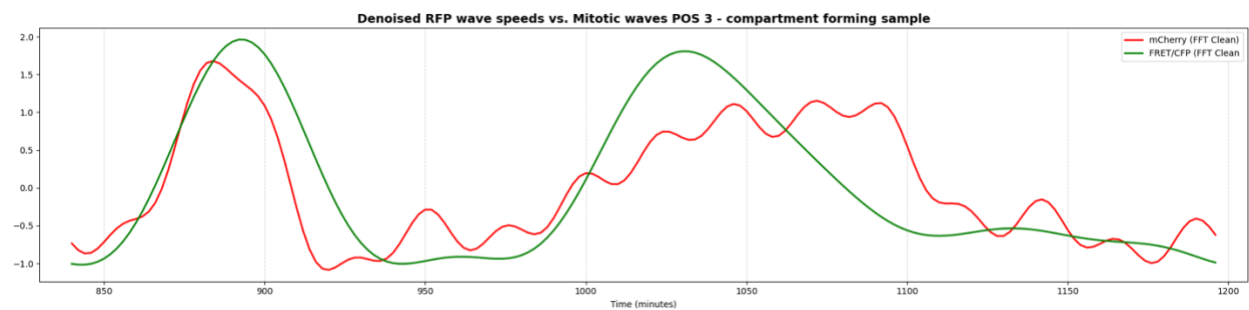
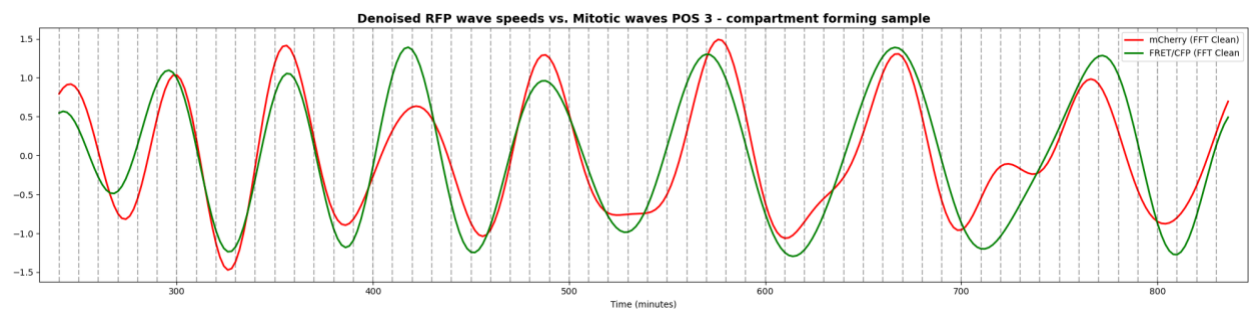
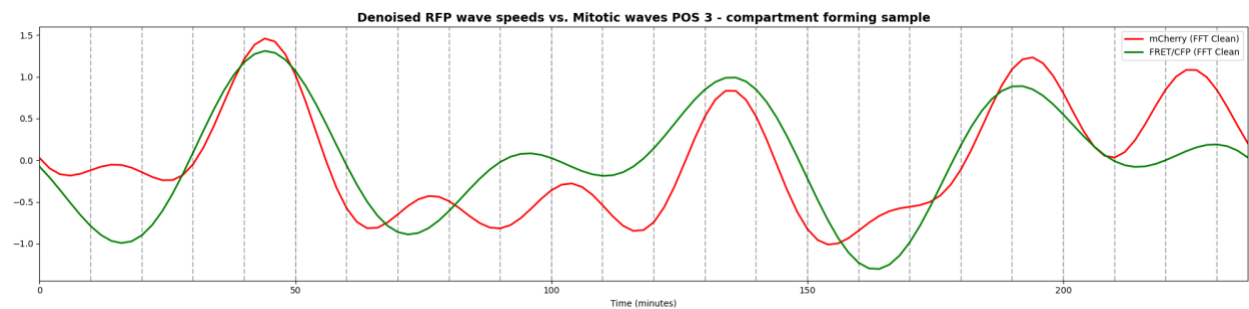
- Assuming that the mitotic activity is lagging relative to the mechanical activity.
- Dynamic viscosity calc → in model 2, this could be improved on.
- Occupancy fraction calc → in model 2, this could be improved on.
- Constant Periodicity, $T = 60$.

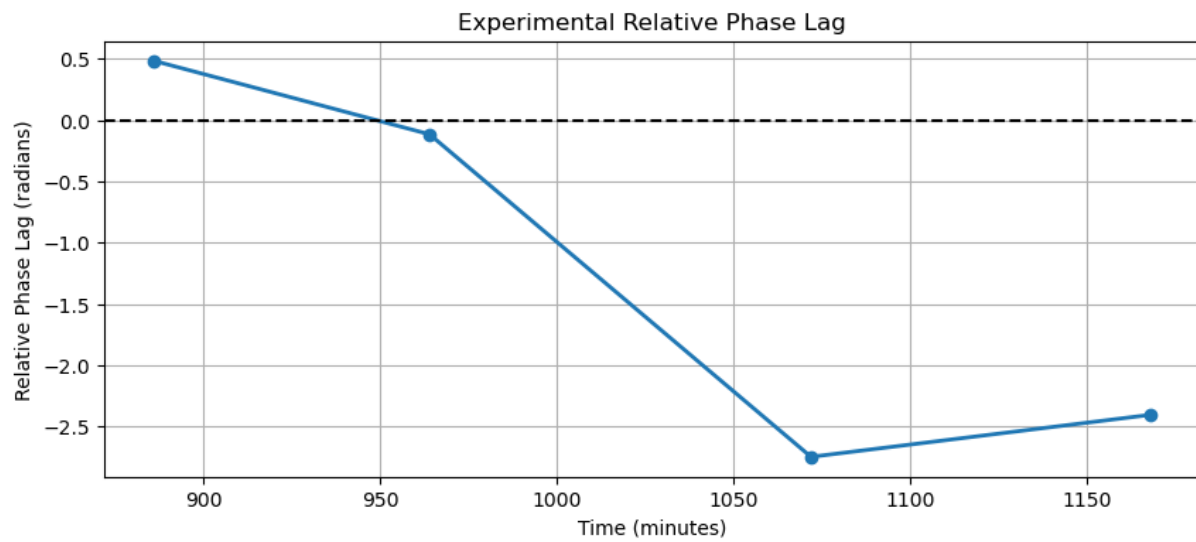
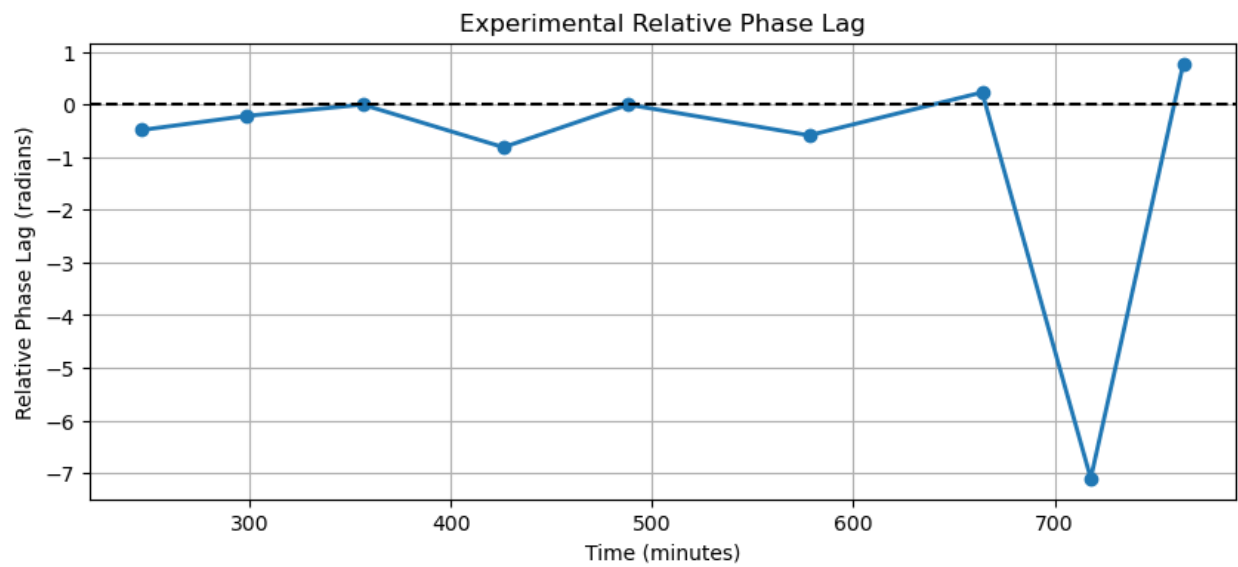
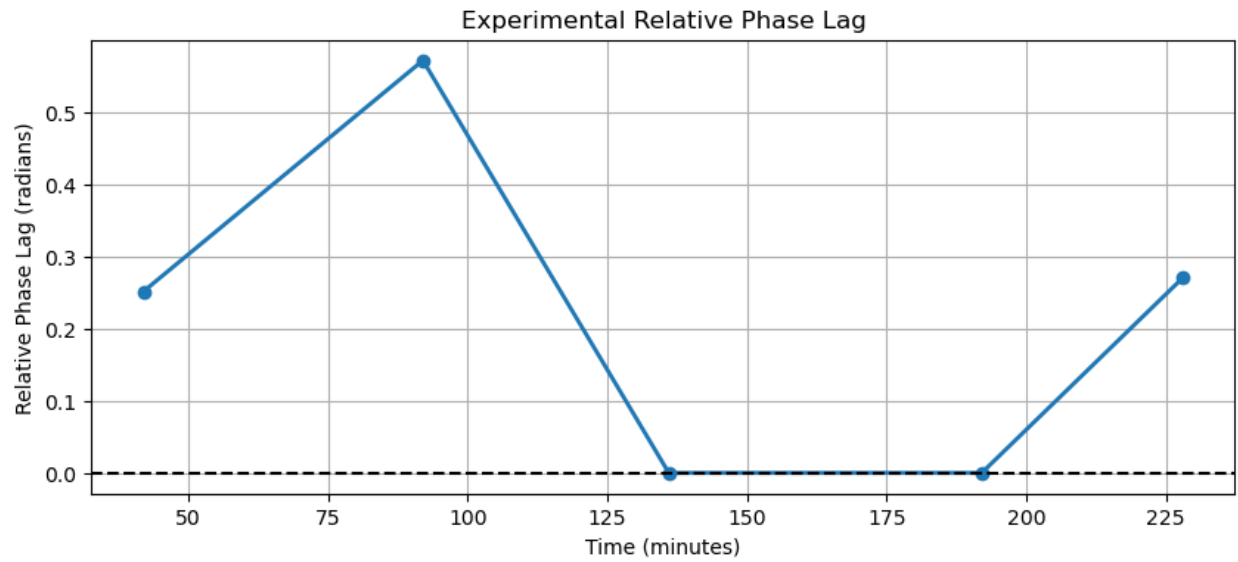
Model Validation:

Compartments:

Model → Experimental



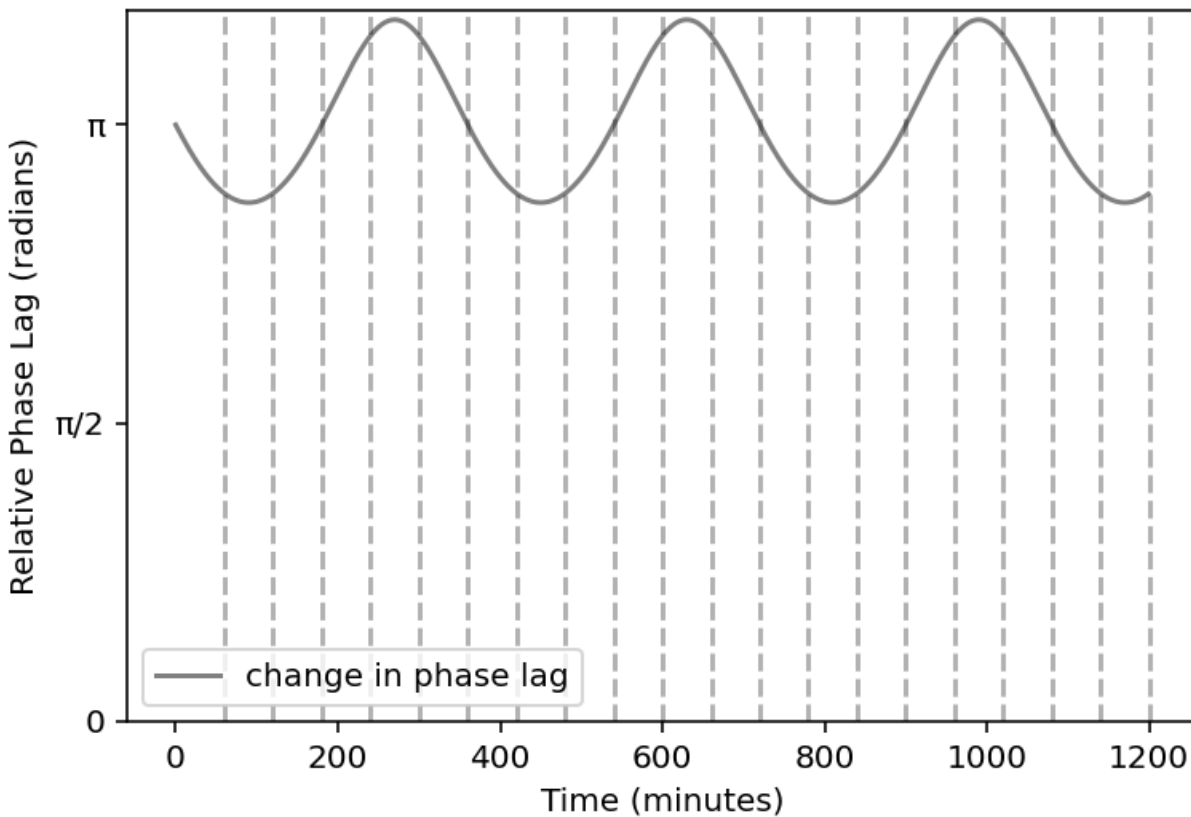


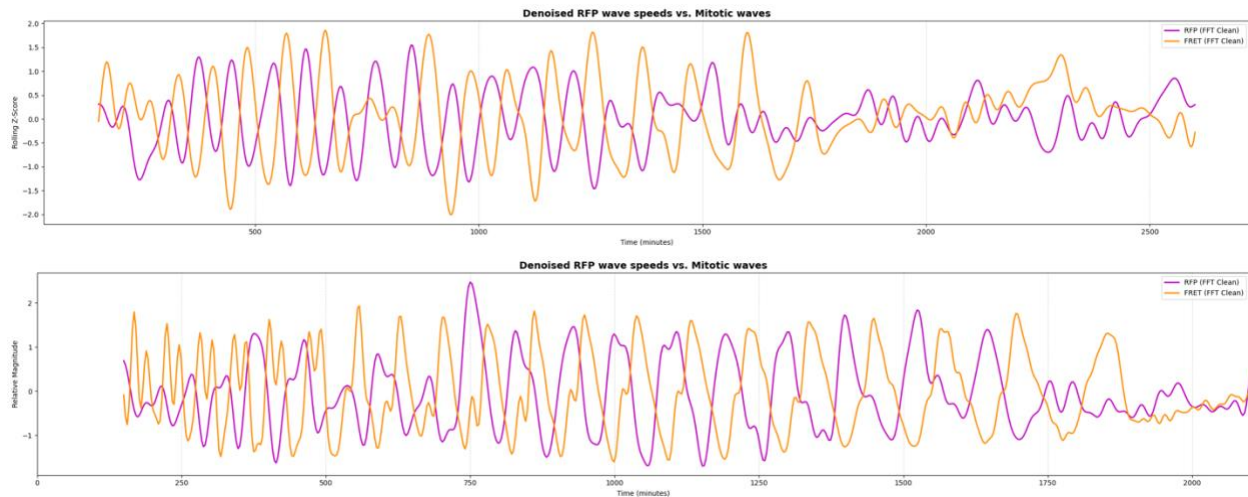


Notes: the compartment forming samples seem to follow a closely inphase relationship. With outliers only occurring when the mechanical activity is variable. The experimental results appear to align with the model results, of course, with more variability. Something interesting is that the mitotic activity begins to lead in the middle of the samples life where mechanical dominates during the other times. We also see the most oscillations per time in the middle chunk of the sample's life. In order to corroborate these results, we need more data (will analyze more positions later!).

No compartment, control:

Model → Experimental





Notes: there is a clear anti-phase relationship here in the non-compartment forming samples. Experimental results look like the model's results. In order to corroborate these results, we need more data.

Other cool questions:

- Does compartment size affect the transport
- FULL ENTROPY AND THERMO MODEL (would be hella sick, free-energy too!)
- Active matter modeling
- Phase-separation, not theory but now test and model
- Multi-compartment wide interactions (do they interact?)

REMEMBER, note to self:

- Trying to not say why compartments form BUT what are the physical ramifications of compartment formation.

Geometry Model

Version 2.0

Model 1.0 vs. Model 2.0:

- After discussing the model with the lab group, there were concerns about the ability to verify whether or not cytoplasm can be classified as a low Reynolds number in this specific system where compartments form due to the appearance of some turbulent flow in microscopy images. This led me to develop a new model where the low-Reynolds number as not assumed and that the turbulence we see in the microscope imagery was caused by the depolymerization of the tubulin clusters. This model varies from model version 1.0 by the fact that the periodicity, wave speed and phase lag all are calculated from experimental data and the idea that the difference between these parameters in compartment forming samples vs non-compartment forming samples is due to the stored mechanical energy released when the tubulin clusters are depolymerized. I feel more confident in this model's results due to the experimental data analysis being more thought through and imposed into this model's calculations.

Big Question:

- What are the downstream effects of the compartment depolymerization events and how do they impact precocity, wave speed and phase shift between 'mitotic' and 'mechanical' activity?

Hypothesis:

- **Tubulin compartment formation stores mechanical energy based on its respective radii; this energy is released as a force (like a spring) when a CDK1 wave hits and depolymerizes the compartment. This outward force affects the 'mixing' or 'mixedness' of the sample, the greater the force, the greater the mixing. This mixing factor is known to affect the rate of transcription. The rate of transcription of cyclin affects the periodicity of the CDK1 wave. Periodicity and wave speed have an indirect relationship but a change in magnitude of wave speed can be derived from the periodicity. Transport can be derived from diffusion rate and wave speeds, which leads to how the phase correlation can shift. Therefore, the compartments act as a mixing agent, which is dependent on their size/geometry, when depolymerized and this event has downstream effects on the systems respective periodicity, wave speed and phase shift between the 'mitotic' and 'mechanical' activity.**

Experimental Observations:

- Sometimes compartments form, sometimes they don't.
- Compartment shape changes throughout the sample's lifetime.
- Sometimes the compartments divide, sometimes they combine.
- The size of the compartment varies over time.
- If left long enough, the compartments will start to crosslink with each other across the regions of no tubulin.

- Wave speed appears to be relatively consistent but then has a very steep drop off prior to arrest.
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- 'Highway' paths of cytoplasm are formed in compartment samples where no tubulin appears to live.
- There is a lot of flow issues in compartment data (in terms of quantifying on ImageJ).
NEW:
- Turbulence appears affect the depolymerization affect
- The compartment size tends to decrease after each cycle, no exactly splitting but...
- The compartments tend to slight darken, intensity drop, prior to the cdk1 hitting. I would not classify this as depolymerization but perhaps an imaging artifact or the compartment density decreasing?
- The shape of the compartment varies as cycles past. In later cycles the compartments are more circular but to crosslink a lot more and turn into 'star' like features.

Assumptions:

- While mixing is not an observable or quantifiable variable, it is assumed that it plays a part in the period length. This relationship is inverse in nature and assumed to be a polynomial like relation, not linear nor exponential.
- We are treating the release of the stored energy as a spring. When the compartment depolymerizes, they act as a spring being released from being stretched. This is due to all of the proteins that get trapped in the cluster due to it's density. When the CDK1 wave hits, the dense cluster of tubulin must now disperse, causing an outward force and carrying the trapped proteins with the force. Prior to the wave hitting there is a compressed and strained potential energy → CDK1 wave → the constraints holding in the energy are removed → the density must disperse (like hand holding a compressed spring being removed)
- Rest Length (of the spring/compartment) = 10 microns
- The max force releasable is normalized to the max force produced during the simulation.
- Our effective spring constant = $1 \text{ e-}6$ (newtons/micron)
- Scaling constant for the wave speed was normalized to the baseline wave speed often observed of 30 microns/minute

- Assuming that the relation between transportation, diffusion and wave speed is direct.
- Assuming that the relation between the relative phase shift and transportation is inverse.
- Assuming that diffusion is a function of mixing and period length.
- Assuming that wave speed is a function of force (therefore also mixing).

Variables:

- **Growth slope** $\rightarrow G_{\text{slope}}(\text{Diameter}, \text{Time})$
- **Stabilization slope** $\rightarrow S_{\text{slope}}(\text{Diameter}, \text{Time})$
- **Polyrate** $\rightarrow \text{Polyrate}(G_{\text{slope}}, \text{Time})$
- **Diameter** $\rightarrow \text{Diameter}(\text{Polyrate})$
- **Max Diameter** $\rightarrow D_{\text{Max}}(G_{\text{slope}}, \text{Time})$
- **Radii** $\rightarrow \text{Radii}(\text{Diameter})$
- **Max Radii** $\rightarrow R_{\text{Max}}(D_{\text{Max}})$
- ΔRadii $\rightarrow \Delta R(\text{Radii}, \text{Time})$
- **Growth Rate** $\rightarrow G_{\text{rate}}(\Delta R, \text{Time})$
- **Stored Mechanical Energy** $\rightarrow E_{\text{Mech}}(\text{Radii})$
- **Released Force** $\rightarrow F_{\text{Released}}(E_{\text{Mech}})$
- **Mixing** $\rightarrow \text{Mixing}(F_{\text{Released}})$
- **Period** $\rightarrow T(G_{\text{rate}}, \text{Mixing}, D_{\text{Max}})$
- **Initial Wave Speed** $\rightarrow \text{CDK1}_{\text{initial}}(\text{Mixing}, T)$
- **Wave Speed** $\rightarrow U(\text{CDK1}_{\text{initial}}, \text{Time})$
- **Initial Diffusion Rate** $\rightarrow \text{Diff}_{\text{initial}}(\text{Mixing})$
- **Diffusion Rate** $\rightarrow \text{Diff}(\text{Time}, \text{Period})$
- **Transport Coefficient** $\rightarrow T_{\text{coe}}(U, \text{Diff})$
- **Phase Shift** $\rightarrow \Delta\phi(T_{\text{coe}})$
- **Minute Lag** $\rightarrow \text{Minlag}(T, \Delta\phi)$

Equations/Parameters & Justifications (in order of the flow of the model):

Units

Equation

'growth_slope' :

- This value was determined by apply the linear line of best fit to the growth portion of the compartment diameter data (“seeing compartment data, sheet 2 & 3 for said data). In the first two cycles, there was not stabilization stage, therefore growth slope was taken over the entire time of the cycle. The growth slope itself is the slope of the linear line of best

fit. There are some caveats with this however, in some situations, the R^2 doesn't fit the line as well as I would like but often the correlation is very good.

'stablization_slope' :

- This value is taken in the same manner as growth_slope. The correlation on average for this varies greatly. In the future I would like to optimize this value, but it does provide the overall trend in terms of producing a compartment size.

'grow_time' :

- This value was determined by the total number of minutes in which the growth phase occurred over. Again, in the first two cycles (0 & 1) the growth time was over the whole cycle, so the growth time is equal to the period length. This data can also be found in the excel sheet. This is in units of **minutes**.

'polyrate' :

- This equation was obtained by looking at either the growth or stabilization phase of each cycle's linear line of best fit where x is equal to the cycle's respective time. 'polyrate' solves for diameter. **$\text{polyrate} = (\text{slope} * \text{cycle_time}) + \text{y-intercept}$**

'effective_diameter' :

- This value is simply derived from the polyrate equation and then translated into microns via the conversion of 2.6 microns/pixel. This value is therefore in the unit's **microns**.
 $\text{effective_diameter} = \text{polyrate} * 2.6$

'effective_radius' :

- This value is self-explanatory. It is in units of **microns**. **$\text{effective_radius} = \text{effective_diameter} / 2$**

'max_diameter' :

- This value is determined via a polynomial equation taken from the experimental data. The max diameter is determined by applying the equation of the 2nd degree polynomial line of best fit to all the max diameter values across all cycles. This is in units of **microns**.
 $\text{max_diameter} = ((1.1517 * (\text{growth_slope}^2)) - (11.414 * \text{growth_slope}) + 586.72) * 2.6$

'max_radii' :

- This value is self-explanatory and in units of **microns**. **$\text{max_radii} = \text{max_diameter} / 2$** .

'delta_radii' :

- This value is self-explanatory and in units of **microns**. **$\text{delta_radii} = \text{curr_radii}[i] - \text{curr_radii}[i-1]$** .

'growth_rate' :

- This value determines the rate of growth a compartment undergoes given the time step. In this case since the time step is 1, $\text{growth_rate} = \text{delta_radii}$. This is in units of **microns/minutes**. $\text{growth_rate} = \text{delta_radii} / \text{dt}$.

'stored_mechanical_energy' :

- The stored mechanical energy treats the growing compartment as if it were a spring being stretched. Therefore, it is in units of **newtons * microns**. The spring constant k_{eff} was determined by taking a spring constant of 1 newton*meters and converting it to 1e-6 newtons * microns. The stored mechanical energy follows a similar equation to a force output of a spring: $\text{stored_mechanical_energy} = 0.5 * k_{\text{eff}} * (\text{radii} - \text{rest_radii})^2$.

'force_released' :

- The force released is proportional to the energy stored but scaling by some constant gamma. This constant would have units of microns^{-1} so that the force could be in **newtons**. $\text{force_released} = \text{gamma} * \text{stored_mechanical_energy}$. For right now, $\text{gamma} = 1.0$ for simplicity.

'mixing_val' :

- How well the system is mixed can be assumed to be related to the force that the compartment releases after the depolymerization event. The force is normalized to the max force output which is determined to be equal to the largest force output of the simulation. The mixing_val is **unitless** and can be considered a coefficient of mixing. $\text{mixing_val} = \text{force_released} / \text{max_force_releasable}$.

'Period' :

- Experimental results have shown that there is a correlation between growth rate and period length and a relation between max diameter and period length.

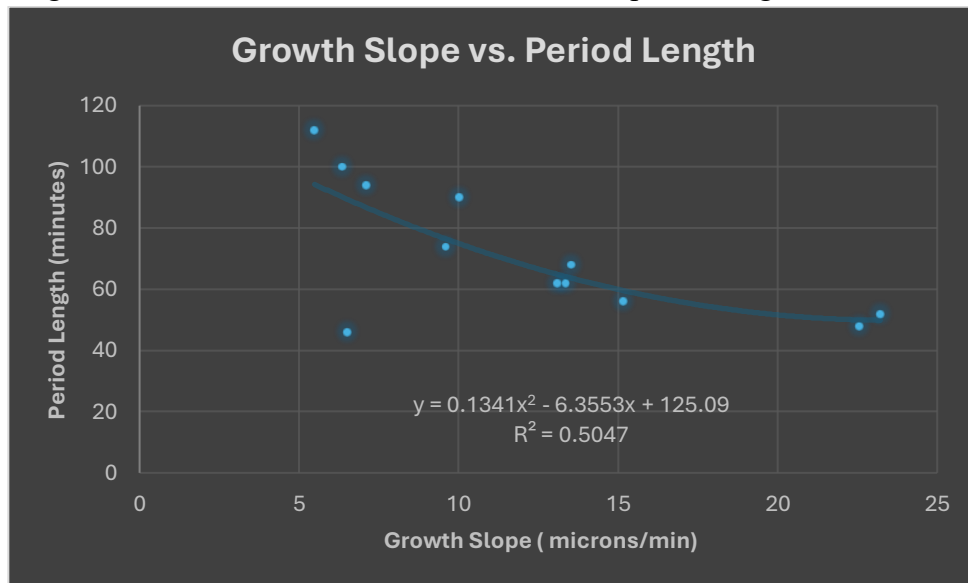


Figure 1. The inverse polynomial relationship between rate of growth and the period length.

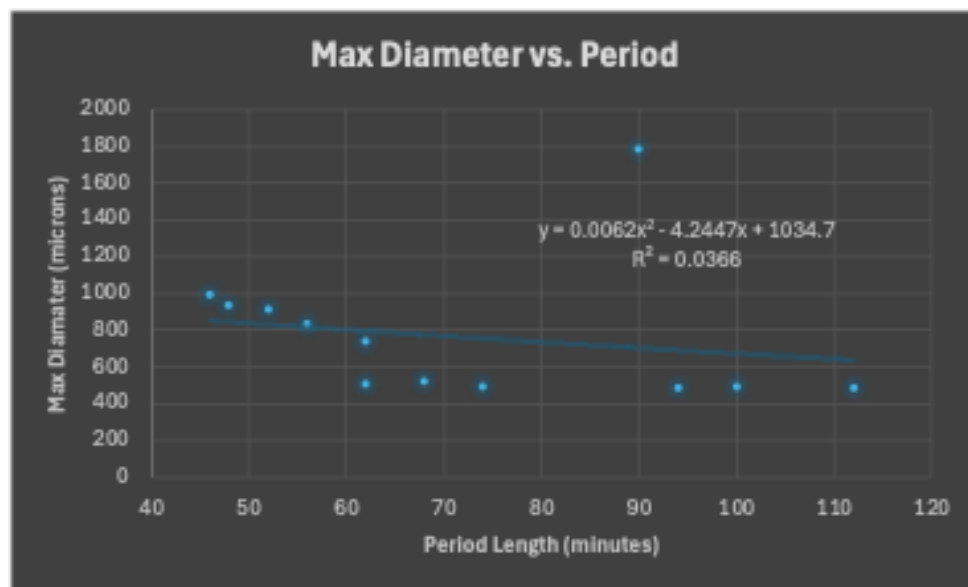


Figure 2. The inverse polynomial relationship between max diameter and the period length (including the max_diameter outlier). This shows a weak correlation. The outlier is most likely due to the cluster combining in cycle one and their max diameter respectively spiking following that behavior.

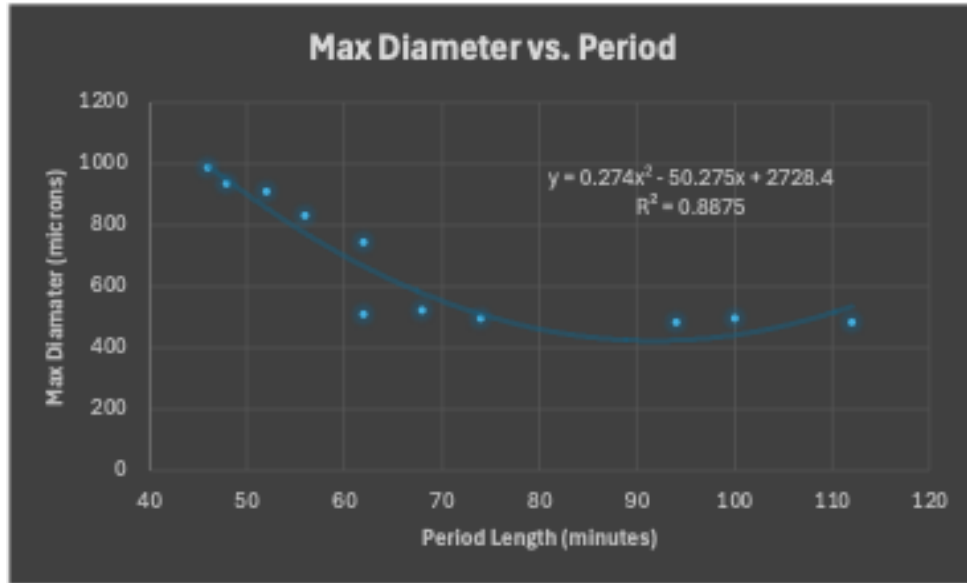


Figure 3. The inverse polynomial relationship between max diameter and the period length (not including the max_diameter outlier). This shows a much stronger correlation.

To quantify the inverse polynomial relationship between period length and max diameter + growth rate, I took the log of both the x & y values to determine the exponent in which the respective parameter would be set to.

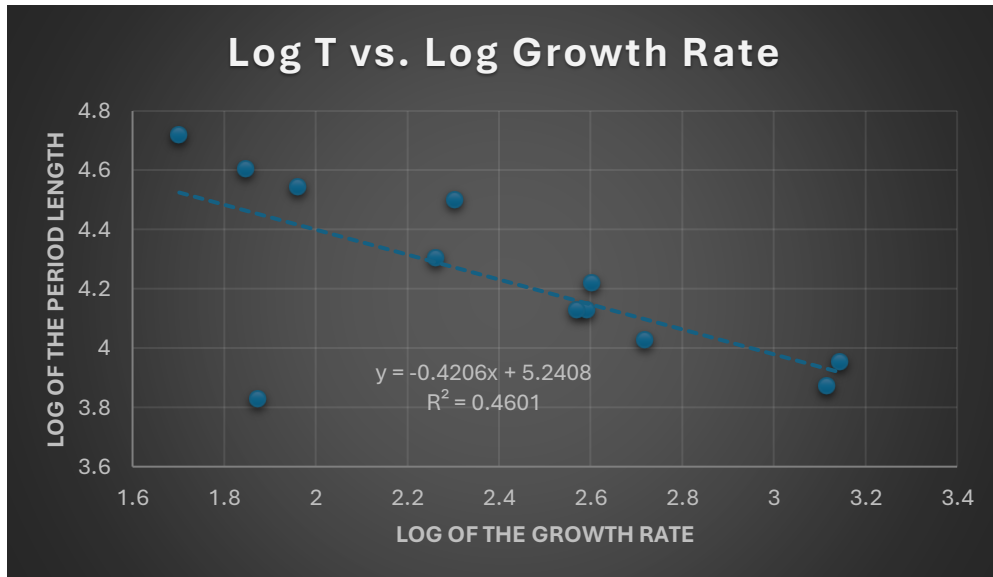


Figure 4. The log vs log relationship between period length and growth rate. The slope of this linear line of best fit is the exponent in which the growth rate will be set to when solving for period length.

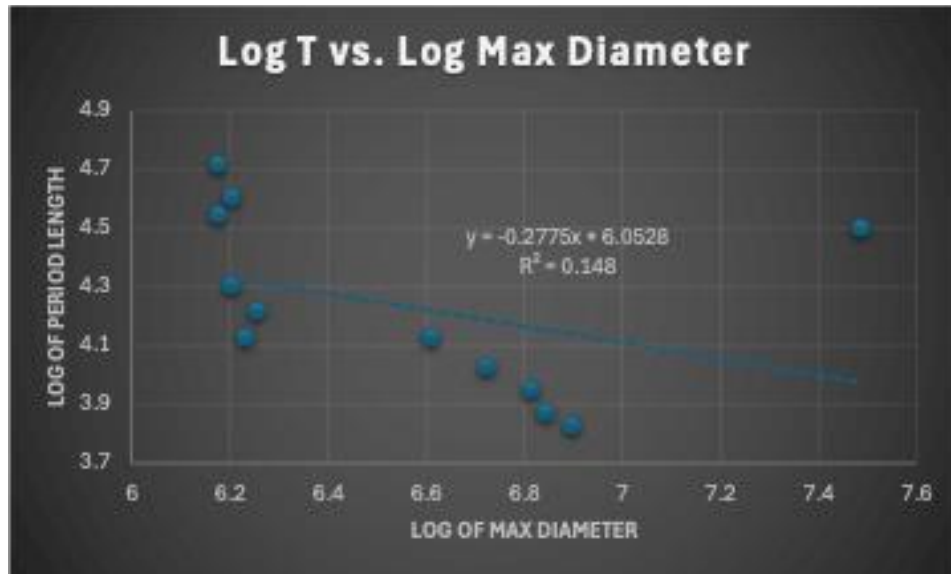


Figure 5. The log vs log relationship between period length and max diameter, including the outlier. The correlation here is weak. The slope of this linear line of best fit is the exponent in which the max diameter parameter will be set to when solving for period length.

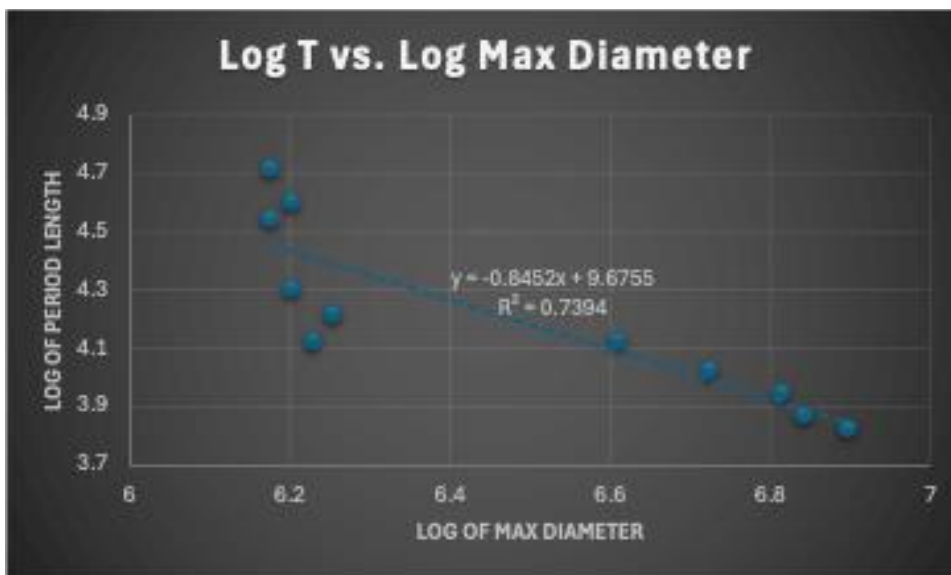


Figure 6. The log vs log relationship between period length and max diameter, NOT including the outlier. The correlation here is much stronger. The slope of this linear line of best fit is the exponent in which the max diameter parameter will be set to when solving for period length.

Since I included the outlier into the model but didn't want the slopes to skew results, I decided to take the mean slope for the log period length vs log max diameter relationship to serve as the exponent, -0.56135 .

Next, I wanted to compare how the change in period length from the initial starting period length related to the cycle. So I created a variable called $\text{ALPHA} = (\text{Period} - T_0) / ((\text{Max_diameter}^{-0.56135}) * (\text{growth_slope}^{-0.4206}))$.

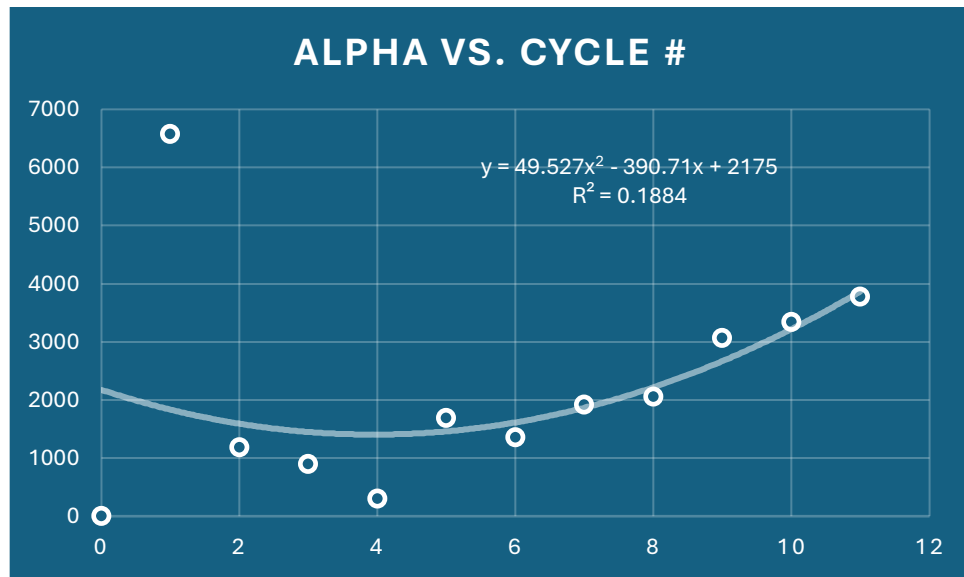


Figure 7. Cycle number vs Alpha, including the outlier, shows a clear trend but poor correlation.

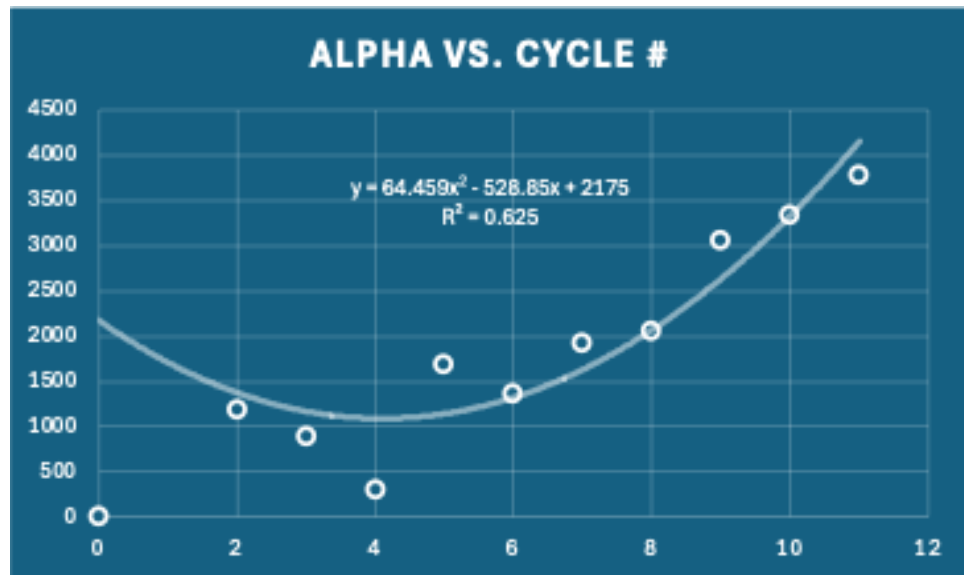


Figure 8. Cycle number vs Alpha, NOT including the outlier, shows a clear trend and better correlation.

The average Alpha value was taken including the outlier and added into the model. $\text{Alpha} = 2175$. To better understand the impact of max diameter and growth rate onto the period variation itself, I compared how the period length changes over time compared to the initial period length vs. $(\text{Max_diameter}^{-0.56135}) * (\text{growth_slope}^{-0.4206})$.

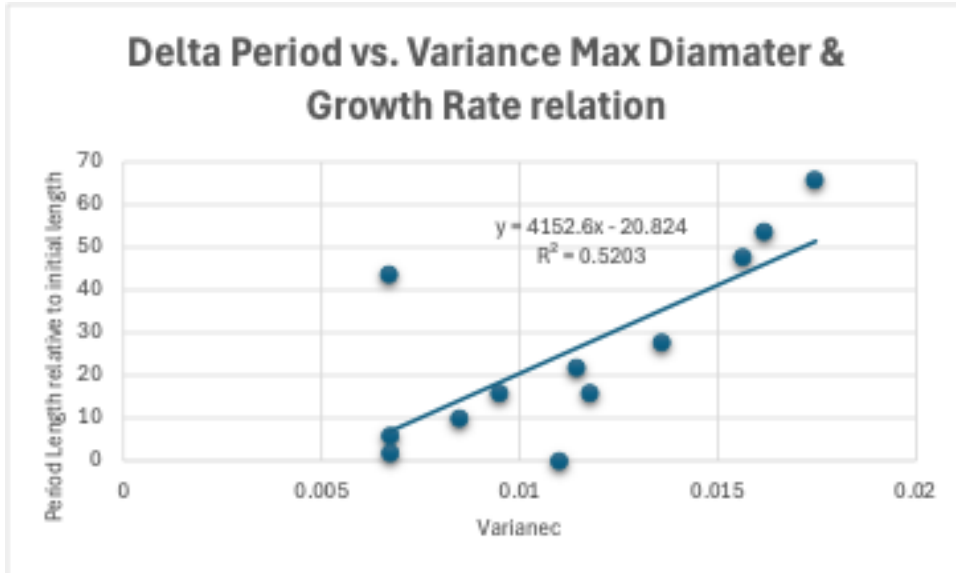


Figure 9. Period length change vs. the variance of max diameter and growth rate, including the outlier. Even when the outlier is included, we see a strong trend and a strong correlation. The correlation implies that about 50% of the period length change is due to the variance in max diameter and the growth rate.

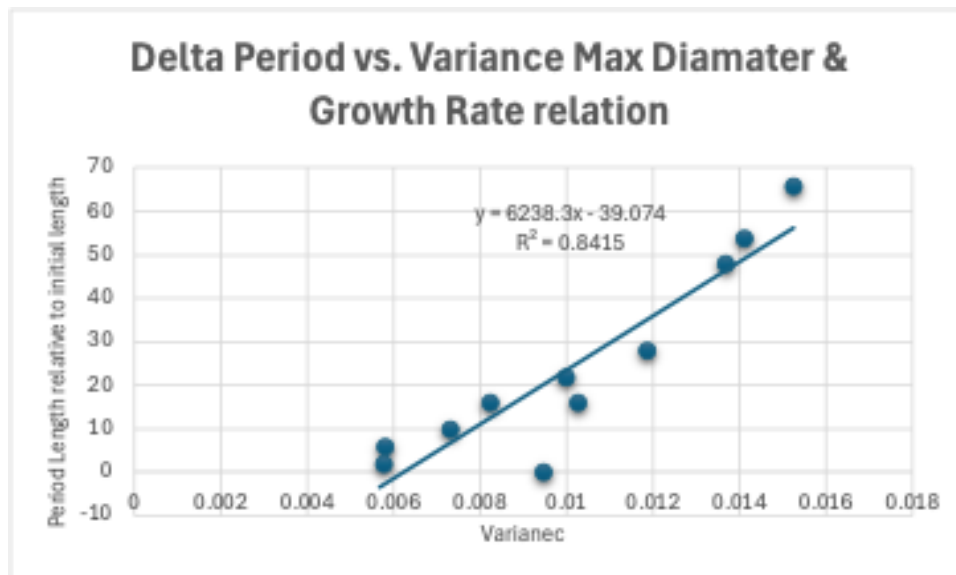


Figure 10. Period length change vs. the variance of max diameter and growth rate, NOT including the outlier. We see a strong trend and a stronger correlation. The correlation implies that about 80% of the period length change is due to the variance in max diameter and the growth rate.

Thus, I assumed that the remaining 50% ~ 20% of the period length would be a result of some mixing variable. Finally, the equation for Period was derived from power-law empirical fits and resulted in:

$\text{Period} = 46 + (\alpha * \text{growth_slope}^{*-0.4206}) * (\text{max_diameter}^{*-0.56135}) * (\text{mixing_val}^{*-theta})$. And is in units of **minutes**. 46 represents the initial period length. Theta is a variable that can be adjusting to fit the experimental results. Currently it sits at 0.65 (50% + 20% = 70% / 2 = 35% - 1 = 65%).

'initial_CDK1_wave_speed' :

- The initial wave speed can be assumed to be a function of both the baseline wave speed (10 microns/minute) and a function of the mixing (which itself is a function of the force). The more well mixed the cytoplasm is, the faster CDK1 signaling proteins can reach each other; therefore, this relationship is proportional. The units are in **microns/minutes**.
 $\text{initial_CDK1_wave_speed} = ((.532 / \text{Period}) * \text{scaling_const}) * (\text{mixing_val} + 1)$. 0.532 is the wavelength for green light in microns, FRET and CFP channels fall around this (I think) and the scaling constant spits out a speed that is within the same order of magnitude as the baseline. $\text{Scaling_const} = 3e3$. The wavelength, baseline speed and scaling_const are all subject to change, they are not set in stone.

'wave_speed_decay' :

- This simply tracks how the wave speed decays throughout the cycle given some drag force. The drag force can also be altered. The units are in **microns/minutes**.
 $\text{wave_speed_decay} = \max(\text{baseline_CDK1_wave_speed}, (\text{initial_CDK1_wave_speed}) - (\text{drag_force} * \text{cycle_time}))$.

'initial_diffusion' :

- This variable is a function mixing and baseline diffusion. We can assume that if the cytoplasm is well mixed, diffusion will increase; therefore, the relationship is proportional. This is in **microns²/min**. $\text{initial_diffusion} = (\text{mixing_val} * \text{baseline_diffusion}) + \text{baseline_diffusion}$.

'diffusion_decay' :

- This variable is self-explanatory. As the tubulin begin to polymerize again, diffusion will be affected, thus the diffusion change is a function of the length of the cycle (period) and what stage of the cycle we are in (cycle_time). This is in **microns²/min**. $\text{diffusion_decay} = \max(\text{baseline_diffusion}, (\text{baseline_diffusion} + (\text{initial_diffusion} - \text{baseline_diffusion}) * \text{np.exp}(-\text{cycle_time} / \text{Period})))$.

'transport_coefiffient' :

- Transport can be assumed to be a function of both diffusion and wave speed. Since transport would increase in both cases of the wave speed increasing and the diffusion rate increase, we can assume that this is a proportional relationship. Both the wave speed and

diffusion are normalized to their baselines. Transport is **unitless**. $\text{Transport_COE} = (\text{wavespeed}/\text{baseline_CDK1_wave_speed}) * (\text{diffusion}/\text{baseline_diffusion})$.

'delta_phase_shift' :

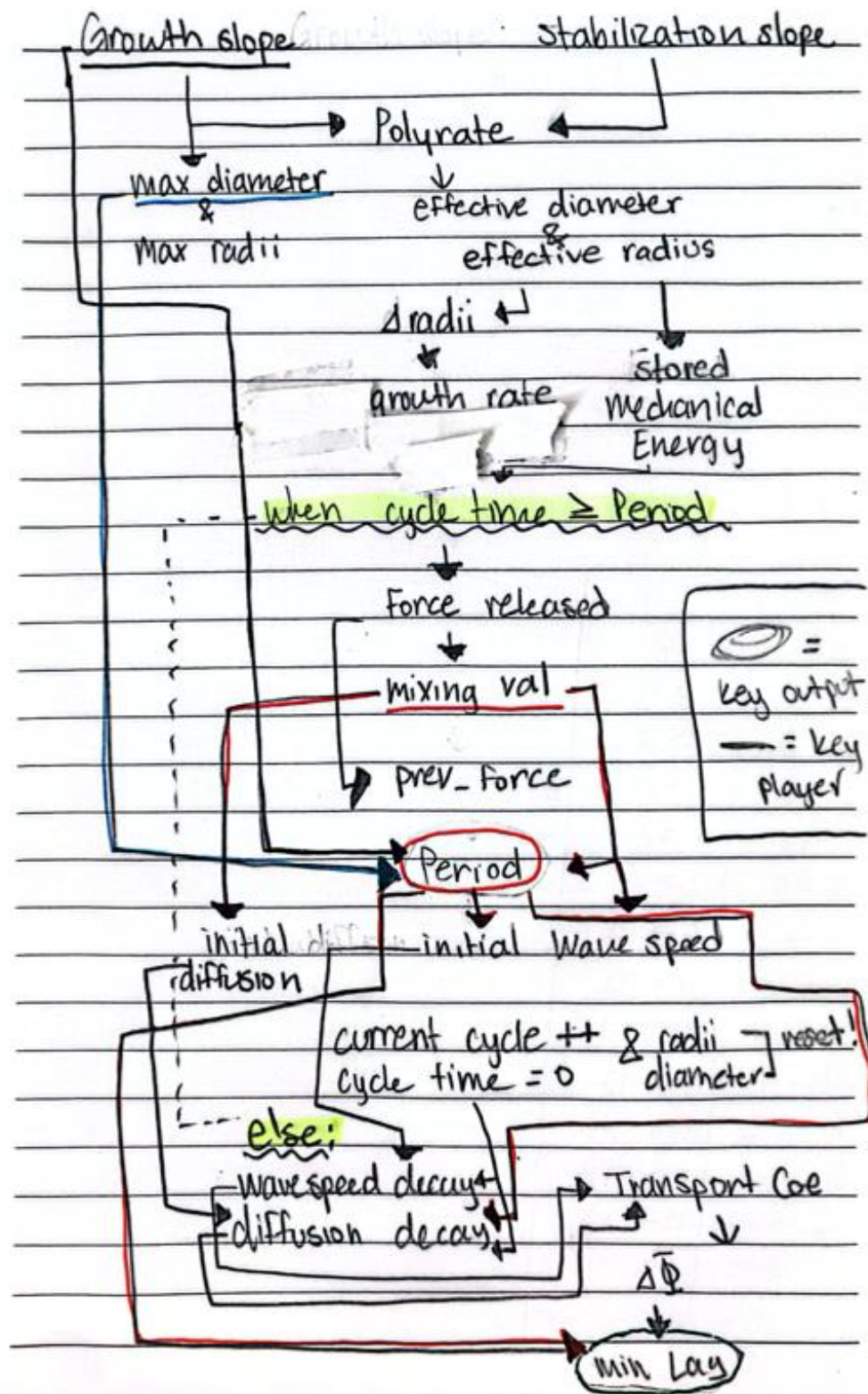
- The change in the phase shift can be assumed to be a function of transportation. If transportation rates decrease there will be more of a delay and vice-versa; therefore, the relationship can be classified as inversely proportional. Delta_phase_shift is **unitless**. $\text{deltaphaseshift} = (1 / \text{transportcoe})$.

'lag_mins' :

- This variable attempts to quantify the lag in minutes between the two activities (mitotic and mechanical). It uses the initial baseline lag from the experimental data to calculate the minute lag based on the phase lag and period. We can assume that the period lag will increase as period length increases and that the phase lag is proportional to the lag in minutes. The lag minutes are in **minutes**. $\text{lag_mins_carrier} = \text{Period} * (8/46) * \text{delta_phase_shift_carrier}$. This way we have a direct value to compare to the experimental results.

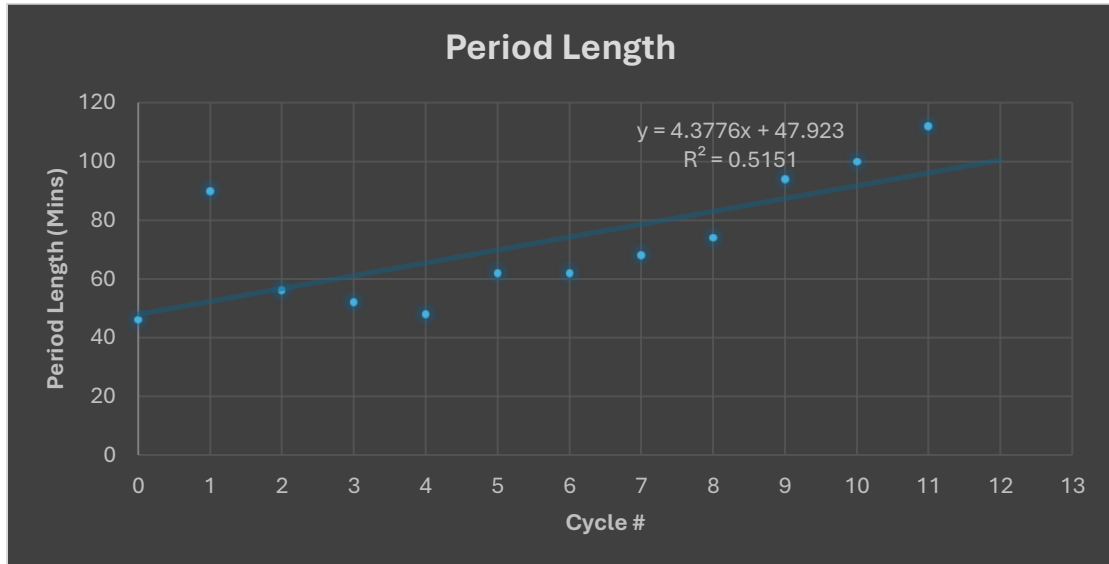
Workflow (how the code is written out):

- Code can be found on GitHub, 'MODEL 2.0'

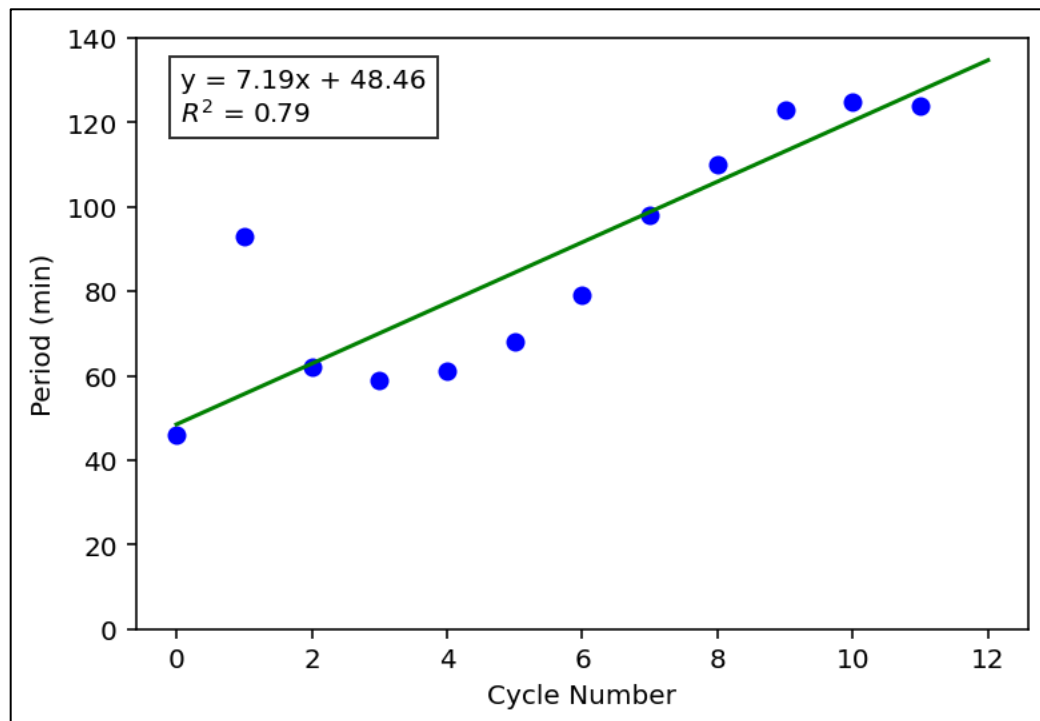


Results:

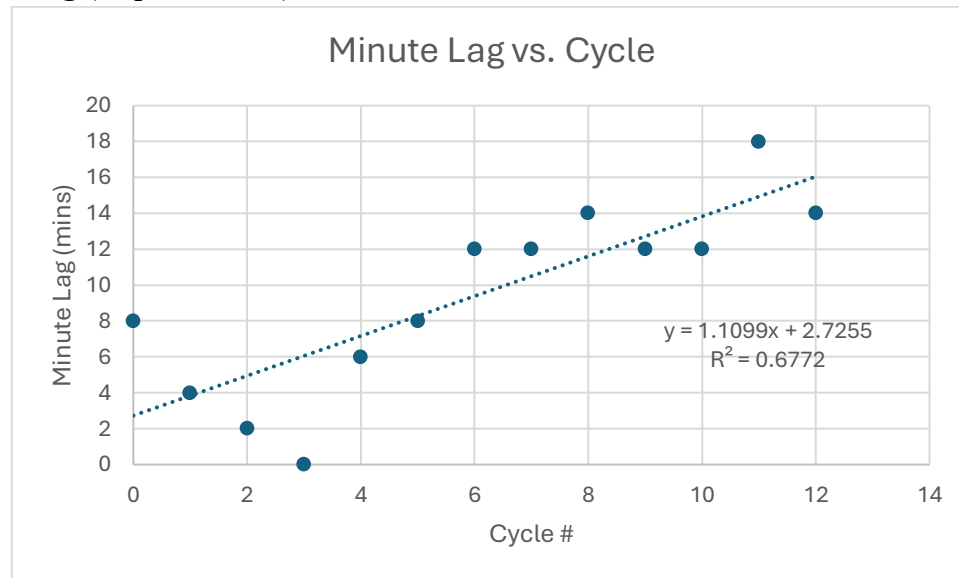
- Period (Experimental):



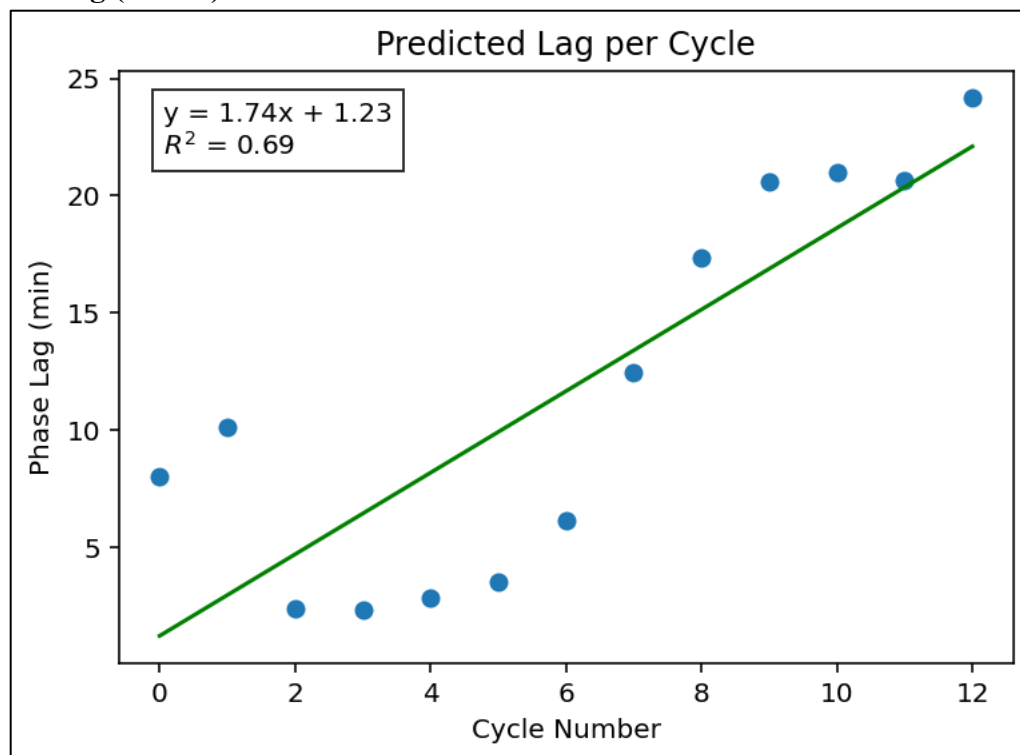
- Period (Model):



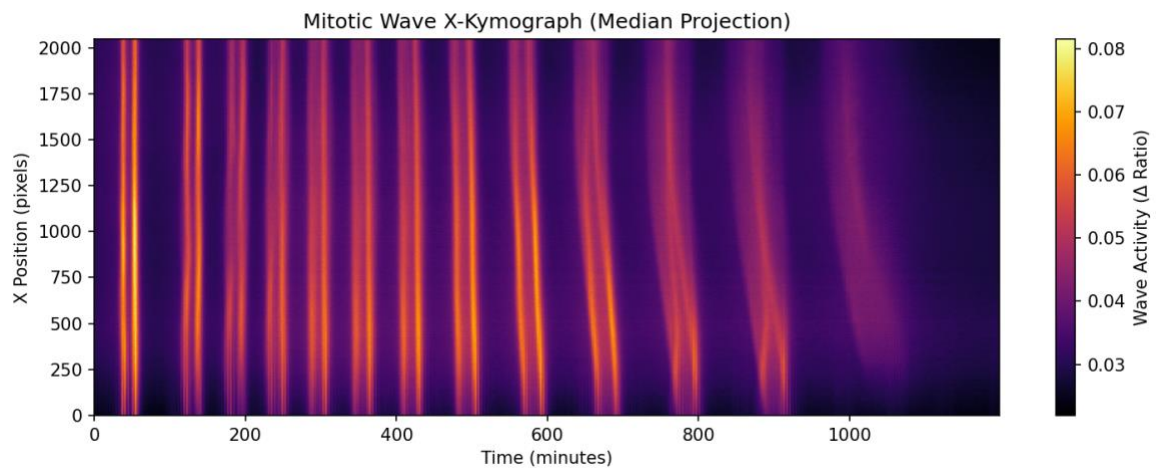
- **Minute Lag (Experimental):**



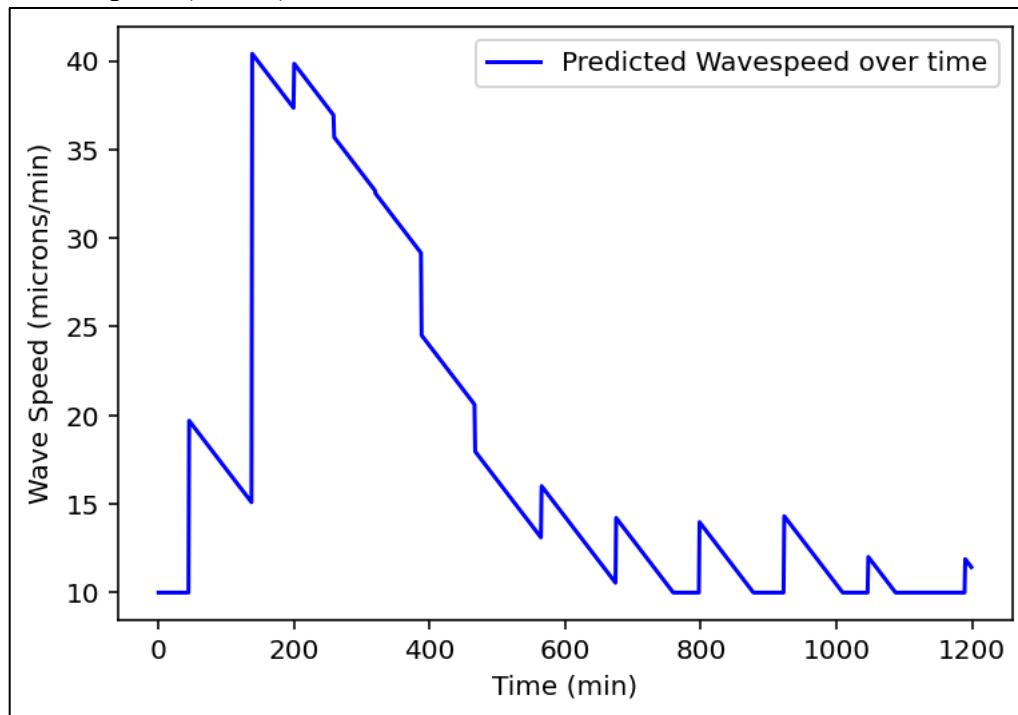
- **Minute Lag (Model):**



- **Wave speed (Experimental):**



- **Wave Speed (Model):**



Discussion:

Overall, the model produces results that have a similar trend to those seen experimentally. Period length appearing to be the most similar. The model's results suggest that the mechanical activity produced by compartment depolymerization affects transport and appears to affect the synchronization of CDK1 events. I developed a framework in the model that links the compartment growth, mechanical activity/energy released to transport-mediated synchronization. This model indicates the geometry of the compartment matters but geometry alone is insufficient. By using the observable compartment growth, the model shows that a mechanistic trend can reproduce the observable period trends seen experimentally. HOWEVER, this is only analysis of one position. Further work should apply these same methodologies to samples that do not form compartments and further samples that do form compartments.

All in all, the trends are not a one-to-one replica, but I think that indicates that there are further components at play here that are yet to analyzed. I think this model provides evidence that period length might be related to mechanical activity within the cell-free extract and the magnitude of said activity plays a role in the variance of period length.

The minute lag results show less of a trend with some discrepancies around cycles 2-5. I believe that the minute lag most likely relies on more than just transport and relative period length. Perhaps the role of wave speed and or viscosity also plays a role.

The wave speed metrics are difficult to quantify experimentally (maybe I will work on this this week??) but the general trend in the kymograph and the model output indicates a stronger wave speed around the 200 – 500 min mark however in the model, we don't see as fast of a wave speed in the first 200 minutes.

Future work should include on fine tuning some of these parameters and equations to get a better understanding as to what mechanisms play into biological oscillatory patterns in *Xenopus* extract. I think understanding the geometry of the compartments, in terms of shape, might play a role in the directionally and the force exerted during the depolymerization event. I also think studying the thermodynamics of this system could be interesting. Noting how slight temperature changes affect the transport could lead to more answers.

Thank you! I had a great summer and a really fun last few weeks working on these two versions. I loved forming a hypothesis and creating the logic for the model and workflow based on the experimental data I had. It was a challenge, but I enjoyed it so much and feel like theoretical and computational physics and biophysics is where I belong! Thank you, UMich Biophysics department, my mentor Lucy, my PI Dr. Yang and everyone else in the Yang lab for helping me out these past 10 weeks. It's been a blast!